This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TEF-CIM: Heterogeneous Tri-Flow Training CIM Macro With Exponent-Exclusive Addend-Compression and Adder-Free Arithmetic Enabling Fused Weight Update

Zhenyu Wang, *Student Member, IEEE*, Yongqiang Mo, Junren Chen, Guoqiang Xin, *Student Member, IEEE*, Wei-Han Yu, *Member, IEEE*, Ka-Fai Un, *Member, IEEE*, Jiawei Xu, *Member, IEEE*, Rui P. Martins, *Life Fellow, IEEE*, and Pui-In Mak, *Fellow, IEEE*

***Abstract*—This article reports a heterogeneous tri-flow compute-in-memory (TEF-CIM) macro that enables end-to-end edge DNN training operations, including forward propagation, error backpropagation (EBP), weight gradient calculation (WGC), and weight update (WU). It features several techniques that enhance the area/energy efficiency and reduce the external memory access: 1) a bit-polarity-interleaved 4T/12T/14T addend-compression matrix-vector-multiplication (MVM) CIM core with complementary 5T-SRAM for hardware-efficient exponent-exclusive 8b (EF8) operation; 2) A vector-outer-product (VOP) CIM core with adder-free bit-select SRAM signed ripple counter, reducing the partial sum access of WGC by 37.3× in ResNet-50; 3) a heterogeneous tri-flow data loading system fusing EBP, WGC, and WU, which reduces the total external memory access by 2.7× in ResNet-50. The proposed TEF-CIM macro prototyped in 28-nm CMOS reaches a peak energy efficiency of 208.6 TFLOPS/W, and an area efficiency of 3.81 TFLOPS/mm^2.**

***Index Terms*—Compute-in-memory (CIM), edge training, exponent-exclusive 8b (EF8), addend-compression, SRAM ripple counter, tri-flow, external memory access, weight update.**

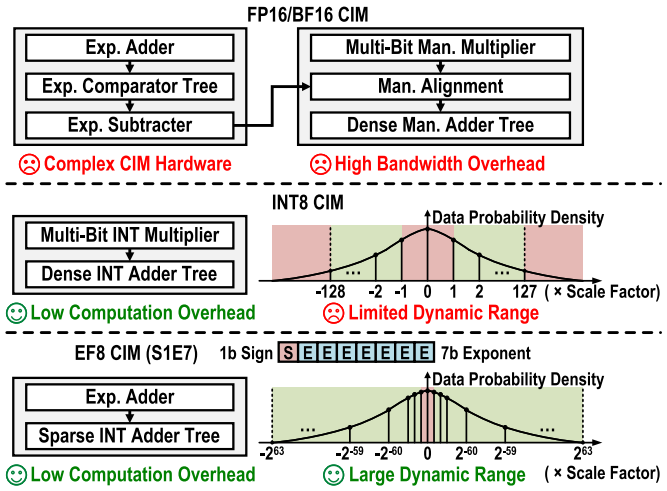


Fig. 1. Comparison of CIM hardware and dynamic range across different data formats.

I. Introduction

DATA-DRIVEN deep neural networks (DNNs) have advanced a broad spectrum of artificial intelligence (AI) applications, including speech recognition [1], image recognition [2], language translation [3], and 3D modeling [4]. Current edge AI devices perform only inference and rely on cloud servers for training, leading to high communication overhead, privacy risks, and inability to adapt to model drift problems in real-world deployment with actual data [5], [6]. Edge training addresses these issues by enabling on-device learning, which preserves data privacy, saves bandwidth, and reduces power consumption in battery-constrained edge devices. Moreover, edge training enables building adaptive and robust DNN models for edge AI devices, while also supporting the localization of transfer learning models [6], [7].

Edge training in typical AI processors requires the execution of forward propagation, error backpropagation (EBP), weight gradient calculation (WGC), optimizer state update, and weight update (WU). The primary data in communication between the processor and the memory involves interlayer activations (IA), weights (W), interlayer errors, weight gradients, optimizer states, and updated weights. Consequently, the data movement overhead in DNN training significantly exceeds the computational overhead in conventional digital processors [8].

Received 23 March 2026; revised 16 June 2026 and 23 July 2026; accepted 31 July 2026. This work was supported in part by Macau Science and Technology Development Fund (FDCT), Macau Special Administrative Region, under Grant 0040/2024/AGJ, Grant 0136/2024/AMJ, Grant 001/2024/COP, and Grant 0001/2025/NRP; in part by the Department of Science and Technology of Guangdong Province (GDSTC) under Project 2025B1212150003; and in part by the National Key Research and Development Program of China under Grant 2024YFE0201100. This article was recommended by Associate Editor L.-D. Van. *(Corresponding author: Wei-Han Yu.)*

The authors are with the State Key Laboratory of Analog and Mixed-Signal VLSI, Institute of Microelectronics, Faculty of Science and Technology, Department of Electrical and Computer Engineering, University of Macau, Macau, China, and also with the Joint Laboratory for Modular Chip Design and Testing, University of Macau Advanced Research Institute in Hengqin, Hengqin, Guangdong 519031, China (e-mail: hankyu@um.edu.mo).

Digital Object Identifier 10.1109/TCSI.2026.3722186

1549-8328 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

By integrating computation into memory, compute-in-memory (CIM) significantly mitigates the training data movement overhead across multilevel memory hierarchies [9], [10], [11]. As shown in Fig. 1, existing floating-point (FP) training CIM employs FP16/BF16 arithmetic, incorporating exponent-domain adders, comparator tree, and subtractors, as well as mantissa-domain multi-bit multipliers, shifters, and dense adder tree [9], [10], [12]. The FP16/BF16 CIM hardware incurs significant I/O bandwidth and computational overhead in DNN training with high-dimensional matrix multiplications. The integer (INT) CIM focuses on fixed-point arithmetic to reduce circuit complexity [8], [13], [14]. Still, the small ratio of the maximum value to the minimum value (i.e., 127) introduces quantization errors for heavy-tailed gradient distributions during backpropagation due to the rounding of small-magnitude values and the clamping of large-magnitude values [15]. It hinders the dynamic range of gradients during backpropagation. The exponent-exclusive data format, derived from logarithmic unbiased quantization, exhibits remarkable data compression from FP format and significantly larger dynamic range than INT format, while maintaining nearly identical training accuracy for CNNs and transformers [15], [16], [17], [18], [19], [20], [21]. Formally, an exponent-exclusive 8b (S1E7) word consists of 1b sign and 7b exponent. With an exponent bias of 63, this format supports a wide dynamic range from 2^{-62} to 2^{63}, and reserves the state of all-zero exponent to represent a numerical value of zero. To the best of our knowledge, the specified training CIM macro targeting exponent-exclusive format has rarely been investigated.

On the other hand, prior CIM architectures employ matrix-vector-multiplication (MVM) dataflow [12], [13], [14], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], which is limited to forward propagation and EBP by the finite multiply-accumulate (MAC) length. Given the significantly higher MAC length of WGC in batch-sample backpropagation, the MVM-optimized CIM is inefficient and requires repeated partial sum access, which typically occupies 97.3% of the weight gradient memory access in ResNet-50 training.

Furthermore, existing MVM-optimized CIM architectures require loading the error matrix from two transposed directions to perform EBP and WGC, incurring repeated memory access [9], [13], [14], [23], which occupies 42% of the overall memory access in ResNet-50. Conventional CIM accelerators also require a separate optimizer to perform optimizer state update and WU [10], [13], [14], resulting in additional DRAM overhead and repeated access of W and optimizer states. To surmount the aforementioned challenges, this paper proposes a 28-nm heterogeneous tri-flow training CIM (TEF-CIM) macro merging all dataflows with hardware-efficient exponent-exclusive 8b (EF8) arithmetic. This TEF-CIM enables efficient processing of forward propagation, EBP, WGC, and WU. The main contributions of this work are as follows.

1) An EF8 bit-polarity-interleaved MVM CIM core for forward propagation and EBP, which leverages 4T/12T/14T addend-compression adders and complementary 5T-SRAM.

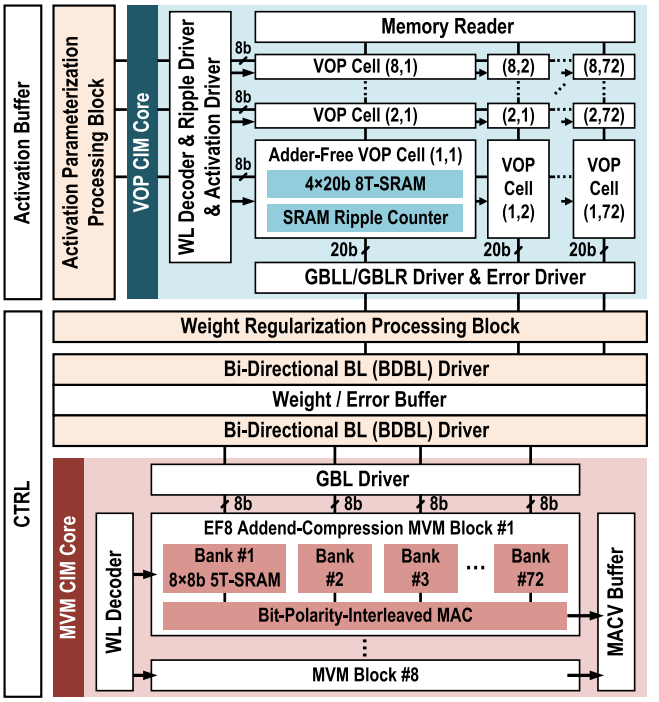


Fig. 2. Overall architecture of the heterogeneous tri-flow training CIM macro.

2) A vector-outer-product (VOP) CIM core featuring the adder-free bit-select SRAM signed ripple counter, which reduces the partial sum access of WGC by 37.3× in ResNet-50;

3) A heterogeneous tri-flow data loading system fusing EBP, WGC, and WU through bi-directional W/error broadcast and activation-error 2D broadcast, which reduces the data access by 2.7× in ResNet-50 training.

The remainder of this article has the following organization. Section II presents the overall architecture of the TEF-CIM macro. Section III details the hardware design of the proposed MVM CIM core and VOP CIM core. Section IV shows the dataflow and data mapping of the heterogeneous tri-flow data loading system. Section V reports the performance and measurement results. Section VI concludes this article.

II. Overall Architecture

The overall architecture of the TEF-CIM is shown in Fig. 2. It consists of a top controller, an MVM CIM core, a VOP CIM core, and data buffers for storing error, IA, and W. The heterogeneous tri-flow data loading system incorporates the bi-directional BL driver, the weight regularization processing block, and the activation parameterization processing block, fusing W writing and EBP/WGC/WU operations of the two CIM cores into a fused framework.

A. MVM CIM Core

The MVM CIM core executes forward propagation and EBP. It contains eight MVM blocks, each with a 72 × 64b

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

5T-SRAM subarray to store the EF8 W. Each MVM block comprises 72 complementary 5T-SRAM banks, and each bank stores eight words of W as a local computing cell for the following exponent-exclusive MAC. Since 3×3 convolution kernels are widely used, 72 banks, which is a multiple of 9, facilitate efficient kernel mapping onto the CIM array. Meanwhile, this bank configuration introduces negligible impact on the data mapping efficiency of fully connected layers and matrix operations in Transformers. The 72×8b GBLs are used to write W into the eight MVM blocks in write mode, and are reused to broadcast IA or error to the blocks for forward propagation or EBP. The WL decoder controls the WLs for write and read of memory. The multiply-accumulate results are sent to the MACV buffer.

B. VOP CIM Core

WGC and WU are processed in the VOP CIM core. The VOP CIM core consists of 72×8 VOP cells, with each cell comprising 4×20b dual-port 8T-SRAM. Every 20b 8T-SRAMs are utilized to store the initial value and updated value in WGC and WU. The data is written into the cells via 72×20b GBLLs and read out via 72×20b GBLRs. The 72×8b errors and 8×8b IAs are broadcast to the cells via horizontal and vertical signal lines for fused WGC and WU. The ripple driver broadcasts ripple signals to trigger the SRAM ripple counter in the VOP cell, and the WLs of the 8T-SRAMs are controlled by the WL decoder. The memory reader precharges the GBLRs and receives the data in the memory under the control of WLs.

III. Hardware Design of CIM Cores

A. Bit-Polarity-Interleaved Addend-Compression MVM Block

Fig. 3 presents the architecture of the MVM block. The EF8 W is stored in the 72 complementary 5T-SRAM banks and is readout to the bit-polarity-interleaved adder via LBLs. The EF8 error is also sent to the adder via GBLs. The bit-polarity-interleaved adder calculates the multiplication of error and W in EF8 format by the addition of exponents and the XOR operation of signs. The max value finder identifies the maximum among all product exponents. The max value is sent to the bit-polarity-interleaved subtractor to calculate the differences between the maximum exponent and all product exponents. These differences and product signs are fed into the exponent-to-one-hot-addend decoder for the following adder tree to compute the summation of the products. The decoder sequentially decodes positive and negative products to avoid signed addition, reducing the toggle rate by 55% in post-layout simulation compared to the 2's complement [32], [34]. The addends are accumulated in the addend-compression adder tree.

1) Complementary 5T-SRAM: To increase memory density, this work proposes the complementary 5T-SRAM. Compared to conventional CIM 6T-SRAM [22], the complementary 5T-SRAM reduces the transistor count by utilizing a single access NMOS transistor. The layout of the complementary 5T-SRAM design features a center-symmetric structure, where two 5T-SRAM cells are

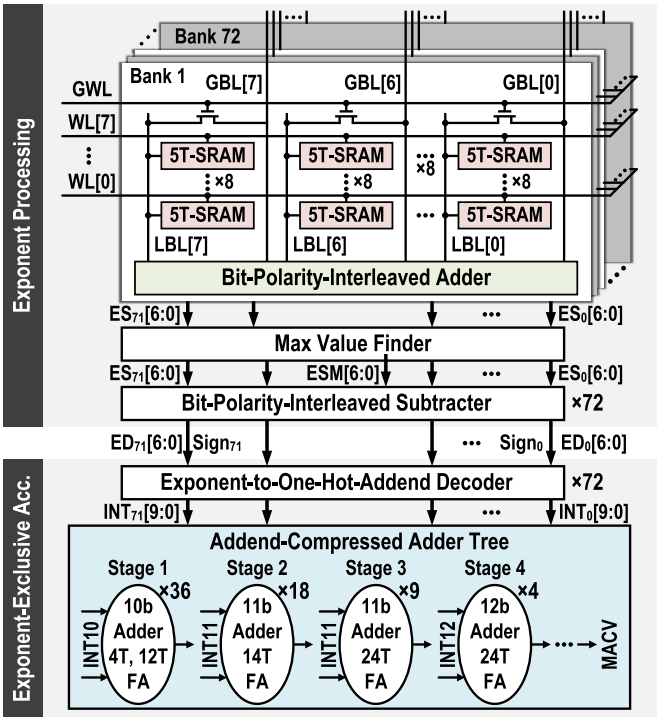


Fig. 3. Architecture of the bit-polarity-interleaved 4T/12T/14T addend-compression EF8 MVM block.

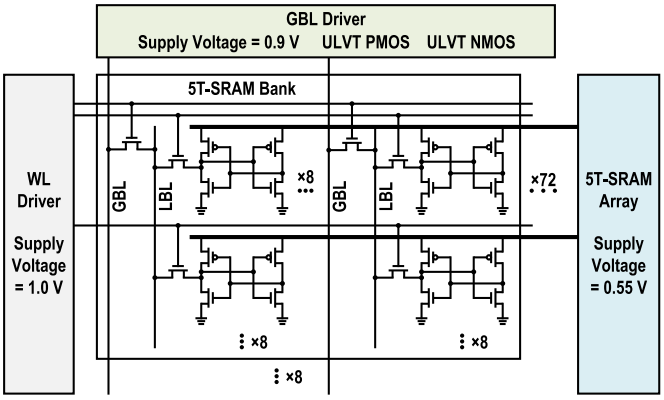


Fig. 4. Diagram of the write scheme of the 5T-SRAM array.

arranged in a mirrored configuration around a central axis. This symmetry allows the layout vacancy resulting from the single access transistor in each 5T-SRAM cell to be effectively compensated by the adjacent 5T-SRAM cell. The complementary 5T-SRAM achieves a bitcell size of 0.29 μm^2 in 28-nm CMOS, representing a more than $1.9\times$ improvement over conventional CIM SRAMs [8], [22], [35]. Fig. 4 illustrates the write scheme of the proposed 5T-SRAM array. The GBL driver operates at a supply voltage of 0.9 V and employs ultra-low-threshold PMOS and NMOS devices to enhance the write driving capability. The WL driver is supplied with 1.0 V to mitigate the voltage drop of the written logic "1" caused by the NMOS access transistor during write operations. In addition, the supply voltage of the 5T-SRAM array is intentionally set lower than that of the GBL driver

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

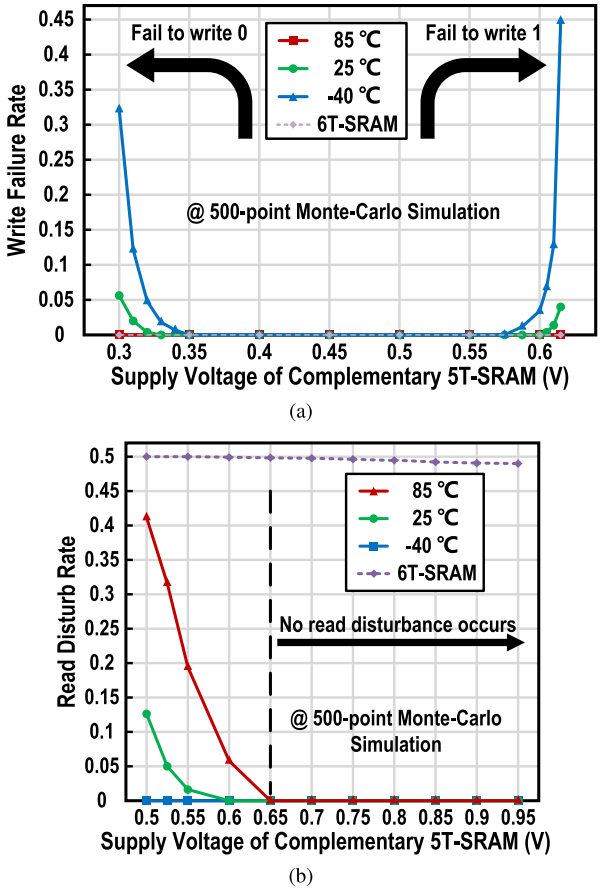


Fig. 5. 500-point Monte Carlo simulation results of the 5T-SRAM array. (a) Simulation results of the write margin under different supply voltages and temperatures. (b) Simulation results of the read disturbance of a local computing cell under different supply voltages and temperatures (The number of WL is 8).

to weaken the latch strength and facilitate successful writes. Fig. 5 shows the 500-point Monte Carlo simulation results of the write and read margin of the 5T-SRAM array under different supply voltages and temperatures. The proposed 5T-SRAM exhibits no write failures over a temperature range from −40 °C to 85 °C under supply voltages ranging from 0.35 V to 0.575 V, and operates without read disturbance over a supply voltage range of 0.65 V to 0.95 V across the same temperature range. Although the conventional 6T SRAM can achieve reliable write functionality over the evaluated supply voltage range, it suffers from severe read disturbance during sequential row readout without BL precharge. Specifically, when adjacent rows store “1” and “0”, and are read successively without precharging the BLs, the read disturb rate is close to 50%. In contrast, the proposed 5T SRAM operates without read disturbance at supply voltage above 0.65 V.

2) Bit-Polarity-Interleaved Addend-Compression Adder Tree: As shown in Fig. 6, the exponent adder adopts the 24T full adder proposed in [36], with 12.5% area reduction compared with conventional 28T full adder [34], [37]. The exponent adder generates interleaved positive-active and negative-active output bits. Due to the sparsity of

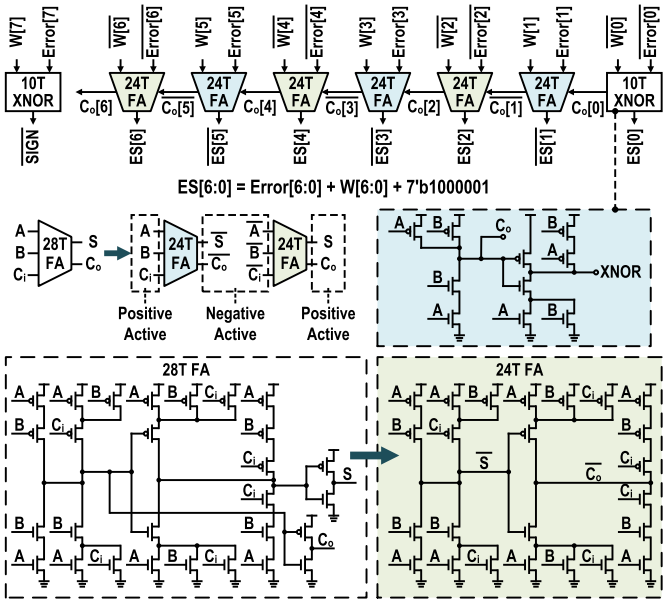


Fig. 6. Diagram of the bit-polarity-interleaved adder with 24T full adders.

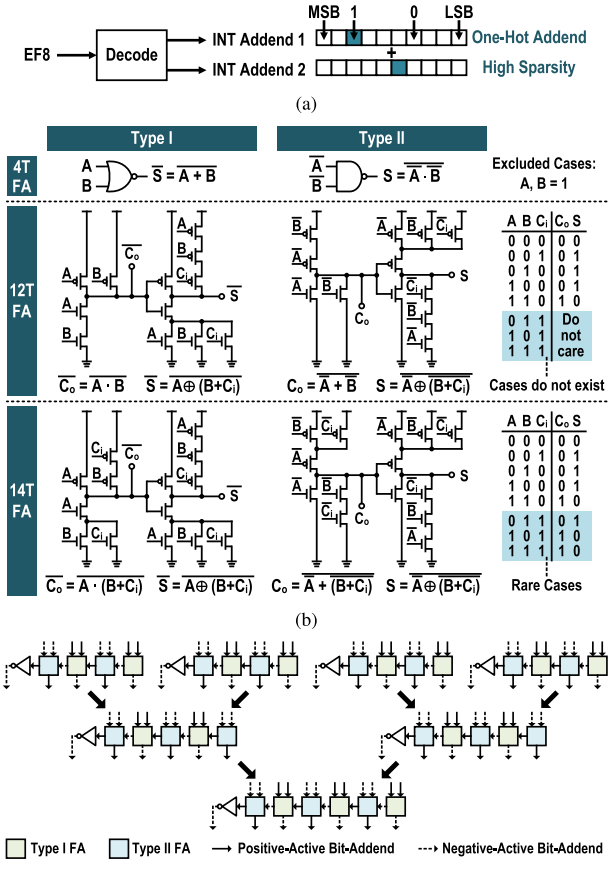


Fig. 7. EF8 bit-polarity-interleaved 4T/12T/14T addend-compression adder tree. (a) High sparsity of exponent-exclusive accumulation. (b) Schematic of the 4T/12T/14T addend-compression full adder (FA) pairs. (c) Diagram of the data transfer across three stages in the adder tree.

exponent-exclusive one-hot addends exceeding 90%, conventional adder trees introduce unnecessary area and energy overhead. The 4T/12T/14T addend-compression adder

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

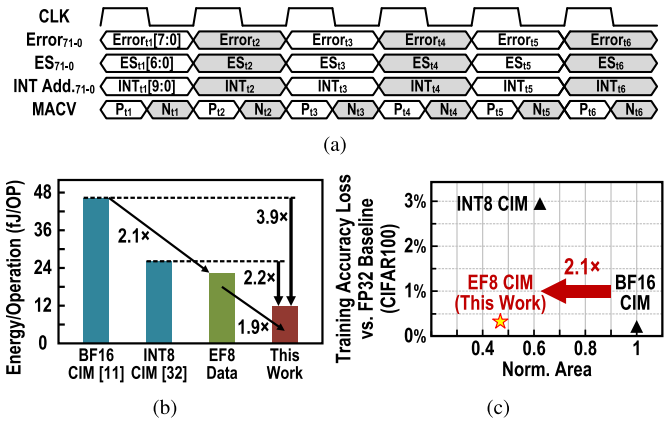


Fig. 8. Timing diagram of MVM CIM block and performance comparison. (a) Timing diagram. (b) Comparison of energy overhead. (b) Comparison of area overhead and training accuracy.

tree is explored for exponent-exclusive formats with only one positive bit. Fig. 3 and Fig. 7(b) illustrate the hardware of the adder tree and the 4T/12T/14T addend-compression full adder pairs based on the bit-polority-interleaved structure, respectively. The type-I full adders in the pairs accept positive-active bit-addends A, B, and carry-in, and generate negative-active sum and carry-out. In contrast, type-II full adders take negative-active inputs A, B, and carry-in, and produce positive-active sum and carry-out. Fig. 7(c) shows the polarity of data transfer across three stages in the bit-polarity-interleaved adder tree. The odd stages receive input data whose least significant bit (LSB) is positive-active and produce output data whose LSB is negative-active. The even stage accepts input data starting with a negative-active LSB and generates output data starting with a positive-active LSB. The first stage of the adder tree is divided into higher 6b and lower 4b. The higher bits are computed with custom bit-polarity-interleaved 12T full adders. The case that carry-in is "1" and one of the bit-addend A, B is not "0" is absent in the 12T adder, since there are only two "1"s in the computation of exponent-exclusive addition. The lower bits are approximately computed with 4T NAND and NOR gates in bit-polarity-interleaved formats, without considering the carry-in and carry-out. The second stage of the adder tree is built with bit-polarity-interleaved 14T full adders. It computes full addition except for the rare case when the carry-in is "1" and both bit-addends are also "1"s. The subsequent stages with fewer adders employ 24T full adders for precise computation. The approximate computation of the adder tree introduces less than 0.1% impact on the training accuracy of ResNet-18. The adder tree reduces the area by 38% and energy by 40% in post-layout simulation compared to the adder tree with all 24T full adders [36].

The timing diagram of the proposed MVM CIM block is shown in Fig. 8(a). Fig. 8(b) presents the simulated energy efficiency comparison between this work and conventional CIM hardware [11], [32]. For fair comparison, we implemented the BF16 CIM [11] and INT8 CIM [32] blocks for simulation. Under the same supply voltage of 0.65 V limited by [32], pseudo-random input bit streams with a 50% toggle

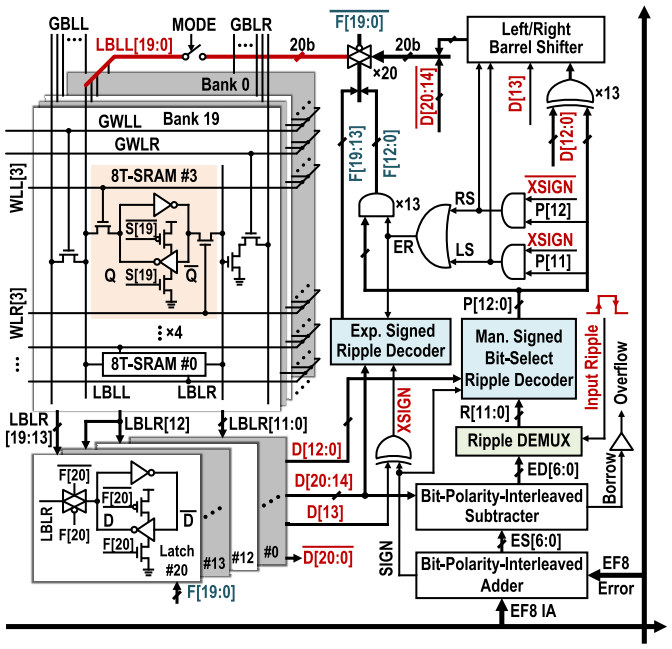


Fig. 9. Diagram of the adder-free bit-select SRAM-signed-ripple-counter VOP cell.

rate were applied, and all designs performed the same 14,400 MAC operations. The average power consumption obtained from circuit simulation was then used to calculate the energy efficiency. Compared with conventional BF16 CIM, the EF8 data format achieves a 2.1× reduction in energy overhead with conventional CIM hardware, and the proposed MVM CIM block provides an additional 1.9× reduction with the optimized hardware. The area overhead is reduced by 2.1× compared with conventional BF16 CIM [11].

B. Adder-Free SRAM-Signed-Ripple-Counter VOP Cell

The VOP cell executes temporal CIM by incorporating an adder-free bit-select SRAM signed ripple counter, which is illustrated in Fig. 9. Each cell incorporates a 20b dual-port 8T-SRAM to store the partial sum Q[19:0], where one port connected to LBLR is for computation, and the other one connected to LBLL is for partial sum update. Within each 20b 8T-SRAMs, the higher 7b store the exponent Q[19:13], and the remaining lower 13b are allocated to store the sign Q[12] and mantissa Q[11:0]. The LBLRs read data into the 21b latches as D[20:0] for MAC. The sign bit is read into two latches as D[13] and D[12], effectively extending the sign bit by one to support the subsequent signed accumulation. The LBLLs operate in two distinct modes. In the write mode (GWLL = 1, MODE = 0), the LBLLs are connected to GBLLs, and the latches within all 8T-SRAMs are unlatched (S[19:0] = 0), allowing data to be directly written into the SRAMs. In the compute mode (GWLL = 0, MODE = 1), the LBLLs and the latches within the SRAMs are controlled by 20b flip flags F[19:0], ensuring that the data of the specific bits which needs to be updated is accurately written into the corresponding SRAMs.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

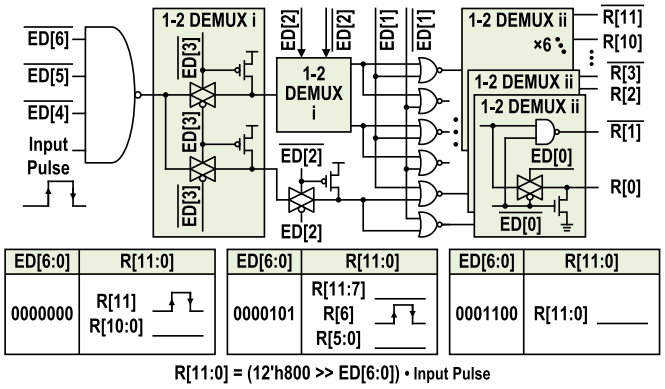


Fig. 10. Schematic of the ripple DEMUX.

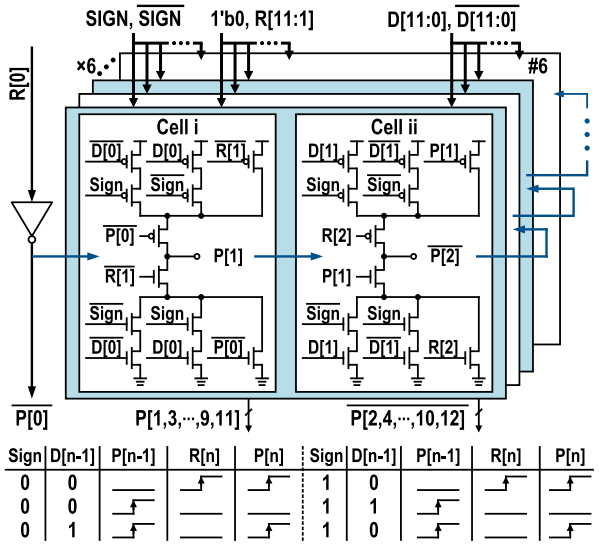


Fig. 11. Diagram of the mantissa signed bit-select ripple decoder.

During each computation cycle, the EF8 error and IA are broadcast to the VOP cell for exponent-domain multiplication. The exponent part of the partial sum D[20:0] stored in the latches is subtracted from the exponent of the product. The difference of exponents and an input digital ripple is sent to the ripple DEMUX as the exponent-exclusive addend and trigger of the SRAM ripple counter, respectively. The ripple DEMUX (Fig. 10) distributes the trigger ripple to one bit of the ripple array R[11:0]. The one active bit of R[11:0] triggers the initial counting bit of the mantissa signed ripple counter.

The mantissa signed bit-select ripple decoder (Fig. 11) generates the sparse digital pulse array P[12:0] based on the one active bit of R[11:0] and the carry/borrow propagation in the counting process. The sign of the addend during accumulation determines whether the ripple decoder propagates the pulse via carry or borrow in the current cycle. The partial sum mantissa flip flags, F[12:0], are derived from P[12:0]. Under the control of F[12:0], the specific bits of the inverted D[12:0] in the latches are connected to the relevant LBLLs. Consequently, the partial sum mantissa bits that need to be flipped during the counting process are updated into the 8T-SRAMs. The exponent signed ripple counter and left/right barrel shifter

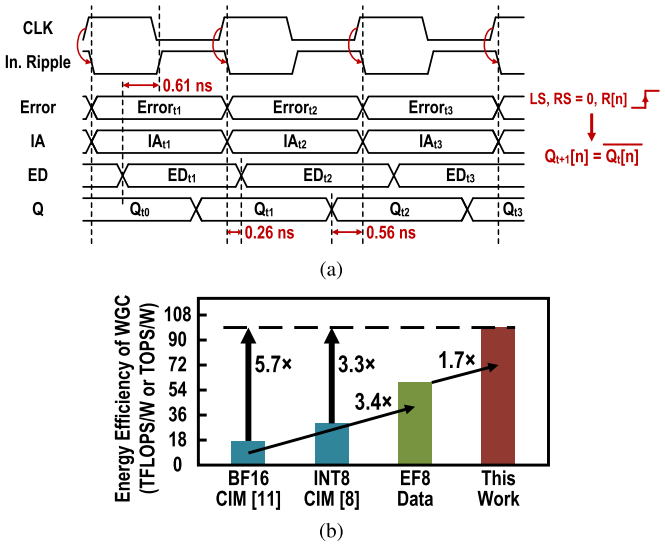


Fig. 12. Simulated timing diagram of the VOP cell and performance comparison. (a) Simulated timing diagram of the the VOP cell operating at 0.95 V and 425 MHz. (b) Comparison of energy efficiency.

are employed for processing normalization of the exponent and mantissa during the accumulation. When normalization is required, the trigger ripple ER for the exponent signed ripple counter is generated by P[12] or P[11]. F[19:13] is generated by the exponent ripple decoder, and F[12:0] is set to all "1"s. The partial sum exponent update is controlled by F[19:13]. The sign/mantissa bits where the up/down counting toggle happens are inverted first, and then all the bits are fed into the left/right barrel shifter for left or right shifting. The shifted data is then fully written into the SRAMs.

The simulated timing diagram of the VOP cell operating at 0.95 V and 425 MHz is shown in Fig. 12(a). The clock signal is broadcast to each VOP cell and locally inverted to generate the input digital ripple for the SRAM ripple counter. The simulation results show that the critical-path setup margins of the computation logic are 0.61 and 0.56 ns during the two half-cycles, respectively. Moreover, the hold margin is 0.26 ns in each computation cycle. These timing margins ensure the correct generation and propagation of the ripple in the SRAM ripple counter, thereby enabling correct memory updates.

Fig. 12(b) shows the simulated comparison of energy efficiency between this work and conventional temporal CIM hardware [8], [11]. The supply voltage is 0.75 V, limited by [8]. Conventional temporal CIM reads the partial sum to the computation logic outside the SRAM array for multi-bit accumulation [8], and the accumulation result is written back into the SRAM array, which incurs significant area and power consumption due to the registers and full adders. The proposed adder-free SRAM signed ripple counter removes the registers and full adders, achieving a 4.4× and 2.5× reduction in area, with energy efficiency improved by 5.7× and 3.3×, respectively, compared to typical register- and full-adder-based BF16 and INT8 CIMs. Compared with conventional BF16 temporal CIM, the EF8 data format achieves a 2.7× improvement in energy efficiency with typical hardware. On top of the in-memory temporal CIM architecture with VOP dataflow.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

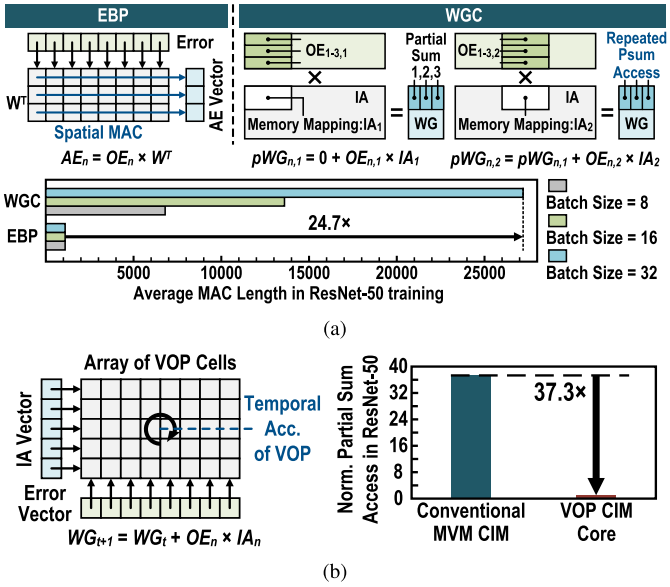


Fig. 13. Comparison of the proposed VOP CIM core with conventional MVM CIM. (a) Conventional MVM dataflow in EBP and WGC. (b) Dataflow of the VOP CIM core and reduced partial sum access.

The CIM hardware also proposes the adder-free bit-select SRAM ripple counter inherently based on SRAM rather than DFFs. The asynchronous ripple propagation exploits the sparse data-flipping behavior of exponent-exclusive accumulation. Compared with the conventional synchronous-clock-based CIM design with full adders and DFFs [37], the SRAM ripple counter improves energy efficiency by 2.1× for EF8 computation.

Different from the near-memory accumulator with large BL capacitance and ripple-selection-based ripple counters [38], the bit-select SRAM ripple counter is inherently based on SRAM rather than DFFs. It reduces the data read/write energy associated with BL capacitance by directly performing computation inside the SRAM array. Furthermore, it supports ripple propagation initiated from an arbitrary bit rather than least significant bit, enabling exponent-exclusive accumulation with a single SRAM ripple counter instead of multiple radix-specific ripple counters. The energy efficiency is increased by more than 1.9× for EF8 in post-layout simulation compared with [38].

Fig. 13 presents the dataflow comparison between the array of VOP cells and conventional MVM CIM. The MVM dataflow is effective for forward propagation and EBP, as EBP represents a transposed process analogous to forward propagation. Different from forward propagation and EBP, WGC involves high-dimensional matrix operations of error and activation. The MAC length increases significantly with larger feature maps and batch size. When the conventional MVM-based CIM architecture is used to perform WGC, the limited MAC length of the memory array allows only an input activation sub-matrix, denoted as IA_n, to be reused in memory. As shown in Fig. 13(a), with IA_1 fixed, only a limited portion of the output error vectors, denoted as $OE_{n,1}$, can be processed to produce partial sum vectors $pWG_{n,1}$, as shown in:

$$pWG_{n,1} = 0 + OE_{n,1} \times IA_{n,1}. \tag{1}$$

These partial sums must be sent off-chip. As the MAC operation proceeds, the reused input activation sub-matrix is updated from IA_1 to IA_2, and another portion of the output error vectors, denoted as $OE_{n,2}$, is processed to generate $pWG_{n,2}$. The previously stored $pWG_{n,1}$ must then be reloaded on-chip, accumulated with $pWG_{n,1}$, and written back again, as shown in:

$$pWG_{n,2} = pWG_{n,1} + OE_{n,2} \times IA_{n,2}. \tag{2}$$

This repeated off-chip access of partial sums is required to complete the full MAC operation in the conventional designs.

In contrast, the proposed VOP CIM core performs WGC through temporal MAC operations over the 2-D array. In each cycle, the outer product between the output error vector OE_n and the input activation vector IA_n is computed and accumulated directly into the partial sum matrix of the weight gradient, as described in:

$$pWG_{t+1} = pWG_t + OE_n \times IA_n. \tag{3}$$

The weight gradient matrix is not written off-chip until the entire MAC computation is completed. As a result, the VOP dataflow reduces the partial-sum access during WGC by 37.3× in ResNet-50 training.

IV. Heterogeneous Tri-Flow Data Loading System

Fig. 14 shows the heterogeneous tri-flow data loading system for fused EBP, WGC, and WU. To fuse EBP and WGC, the error is bi-directionally broadcast to MVM and VOP CIM cores via the bi-directional BL driver. The IA vector is broadcast in a direction orthogonal to the error vector in the VOP CIM core to facilitate WGC. Conventional MVM-optimized CIM only performs spatial MAC. During EBP and WGC, the error vectors are loaded from two transposed directions, resulting in repeated access of the same error matrix. The fused EBP and WGC reuse the same error vector to perform spatial MAC for EBP and temporal MAC for WGC, thereby eliminating repeated data access.

Moreover, we propose in-memory WU, which directly utilizes the products during the MAC operation of WGC to update the partial sum as W with the dual-port 8T-SRAM. Consequently, WGC and WU are fused automatically in the VOP CIM core. Before being sent to the VOP CIM core, the IA is multiplied by the negative learning rate $-lr$, which is also in EF8 format, introducing minimal computational overhead. Conventional hardware executes WGC, optimizer state update, and WU in the separate training core and optimizer, as shown in Fig. 14(c), which leads to repeated W and optimizer state access. The fused WGC and WU dataflow reduces the repeated access of W and optimizer states and off-chip DRAM overhead.

As shown in Fig. 14(d), similar to the training dataflow of IA–W matrix multiplication in CNNs, the training operations of Q-K^T matrix multiplication in the attention score layer include EBP of Q and K^T. The MVM CIM core computes the Q error vector from the output error vector and the K^T matrix,

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

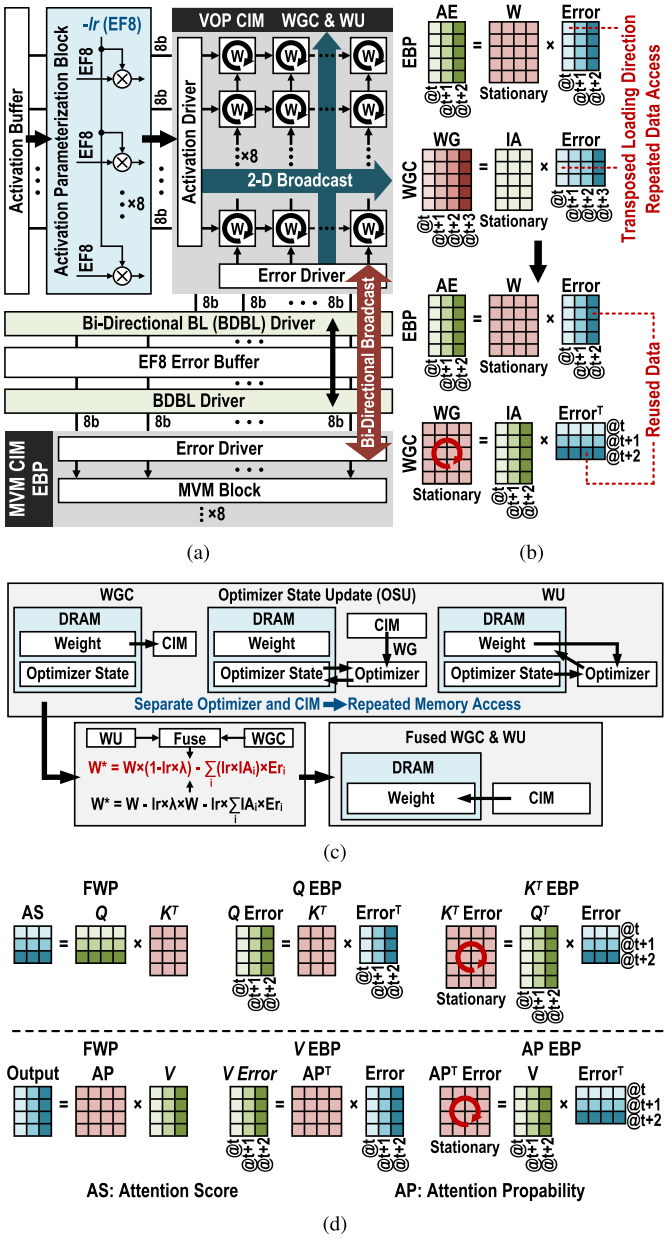


Fig. 14. Heterogeneous tri-flow data loading system for fused EBP, WGC, and WU. (a) Diagram of fused EBP, WGC, and WU. (b) Comparison of conventional CIM and fused EBP and WGC. (c) Diagram of the fused WGC and WU. (d) Dataflow of the training operations for the attention score layer and the attention output layer in the transformer.

while the VOP CIM core accumulates the outer products of the output error vector and the Q vector to obtain the K^T error matrix. The training operations of attention–V matrix multiplication in the attention output layer include EBP of V and attention probability. The MVM CIM core computes the V error vector from the output error vector and the attention probability matrix, whereas the VOP CIM core accumulates the outer products of the output error vector and the V vector to calculate the attention probability error matrix. Since no WU operation is required in either case, the memory cells in the VOP CIM core are initialized to zero before computation.

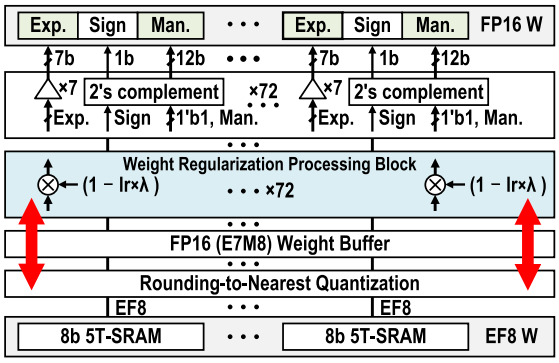


Fig. 15. Diagram of the weight loading mechanism across two CIM cores.

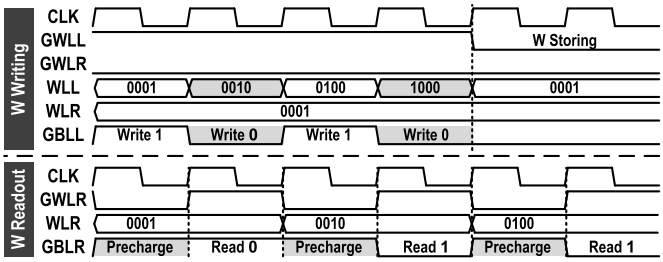


Fig. 16. Timing diagram of the weight loading and readout.

The training dataflow of QKV projection layers is the same as that of CNNs.

Fig. 15 shows the weight loading mechanism across two CIM cores. The W is stored in 8T-SRAM with FP16 (E7M8) format to ensure training precision for WU, and is quantized to EF8 to fuse with EBP. The training macro also supports training regularization. The W is multiplied by $(1-lr \times \lambda)$ to address L2 regularization during model training, where λ represents the regularization hyperparameter. The processed W is written into the 8T-SRAM as the initial values for WU. The trained W is read off chip via GBLRs. Fig. 16 illustrates the timing diagram of the weight writing and readout. The data loading system eliminates redundant W and error access in conventional training, achieving a 2.7× reduction in memory access.

V. Measurement Results

Fig. 17 presents the micrograph of the proposed TEF-CIM macro fabricated in a 28-nm CMOS process, which occupies a chip area of 1.4×0.9 mm^2, including the pads. The measurement setup of the TEF-CIM is demonstrated in Fig. 18. A signal generator is utilized to generate the clock signal for the test chip. An FPGA generates the required IAs/Ws/errors and processes the output results from the test chip. An oscilloscope shows the computed results for debugging.

Table I shows the chip summary. The TEF-CIM macro contains complementary 5T-SRAM for EBP and dual-port 8T-SRAM for fused WGC and WU. The matrix multiplication is performed using EF8 precision for both IA and errors throughout training. The W is processed in EF8 format during EBP, while retaining full FP16 precision during WU. Under 0.55-0.95 V supply voltage, the working frequency is

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

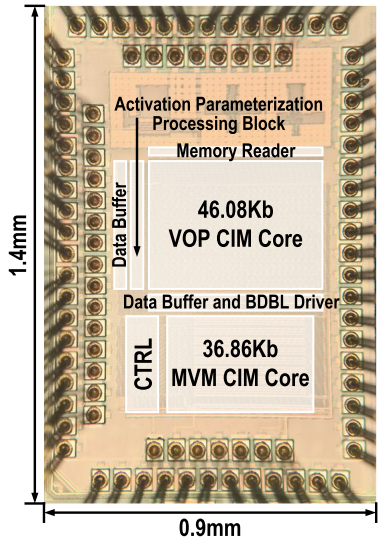


Fig. 17. Chip micrograph of the heterogeneous tri-flow training CIM macro.

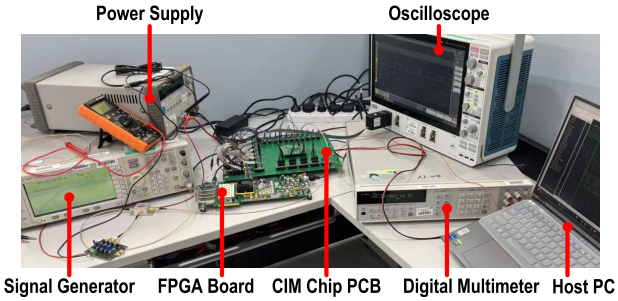


Fig. 18. Measurement setup of the CIM macro.

TABLE I
CHIP SUMMARY

Technology	28nm CMOS
Die Area (mm^2)	1.26
Macro Size	82.944Kb
Error Precision	EF8
Activation Precision	EF8
Weight Precision	EF8 (EBP), FP16 (WU)
Voltage (V)	0.55 - 0.95
Frequency (MHz)	110 - 425
Access Time	2.5ns @0.9V
Area Efficiency (TFLOPS/mm^2)	0.98 - 3.81 (EBP & WGC & WU)
Average Energy Efficiency [a] (TFLOPS/W)	33.1 - 108.5 (EBP & WGC & WU)
Peak Energy Efficiency [b] (TFLOPS/W)	74.5 - 208.6 (EBP & WGC & WU)
Training Accuracy [c]	91.60% [d] @CIFAR10 (ResNet-20)
	77.48% [d] @CIFAR100 (ResNet-18)
	76.04% [d] @ImageNet (ResNet-50)
	88.13% [e] @SQuAD (BERT-base)

[a] Measured with 50% error/activation/weight sparsity
[b] Measured with 50% weight sparsity and 10% toggle rate
[c] Software FP32 baselines of 4 models are 91.80%, 77.76%, 76.38%, 88.36%
[d] Models are trained from scratch
[e] Model is trained from pre-trained BERT-base

110-425 MHz across 4 testing chips. The TEF-CIM macro achieves a state-of-the-art (SOTA) average energy efficiency

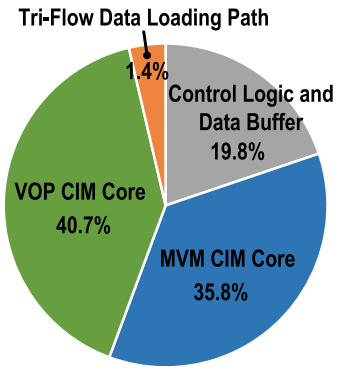


Fig. 19. Energy breakdown of the chip.

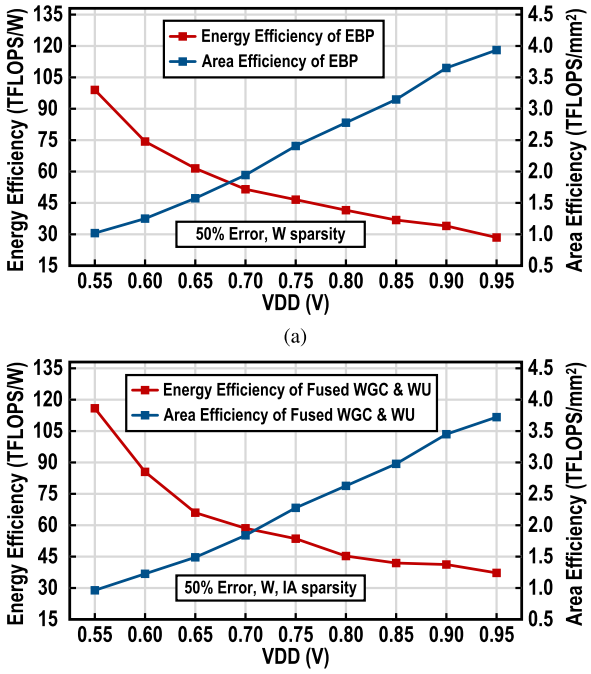


Fig. 20. Energy/area efficiency voltage scaling. (a) Energy/area efficiency voltage scaling of EBP operation in the MVM CIM core. (b) Energy/area efficiency voltage scaling of fused WGC and WU operation in the VOP CIM core.

of 33.1-108.2 TFLOPS/W for fused EBP, WGC, and WU. The peak energy efficiency of the TEF-CIM macro is 208.6 TFLOPS/W at 0.55 V and 110 MHz, and the peak area efficiency is 3.81 TFLOPS/mm^2 at 0.95 V and 425 MHz for fused EBP, WGC, and WU.

Fig. 19 depicts the energy breakdown of the chip. Due to the two-dimensional temporal computation in the VOP CIM core, which requires processing high-bitwidth partial sums, the energy consumption of the VOP CIM core for WGC is higher than that of the MVM CIM core for EBP. In contrast, the tri-flow data loading path incurs substantially lower energy overhead than two CIM cores, as the multiplication between the learning rate and IA in WU is implemented using exponent addition, and the broadcast path is short.

Fig. 20 illustrates the measured energy and area efficiency of two CIM cores under voltage scaling from 0.55 V

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

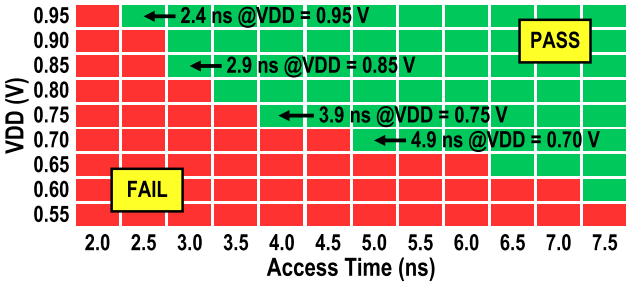


Fig. 21. Measured shmoo plot under different VDD: EF8 Error, EF8 W (MVM CIM), FP16 (VOP CIM), EF8 IA.

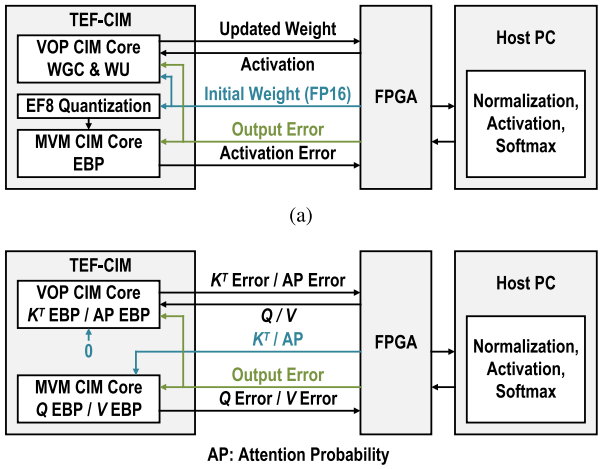


Fig. 22. Diagram of the end-to-end training platform. (a) Training flow for convolution and linear layers. (b) Training flow for the attention score layer and attention output layer.

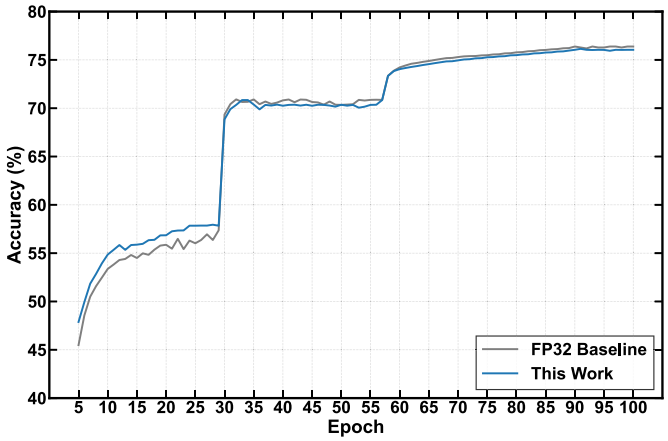


Fig. 23. Training accuracy curve of the TEF-CIM based training platform for ResNet-50 on ImageNet.

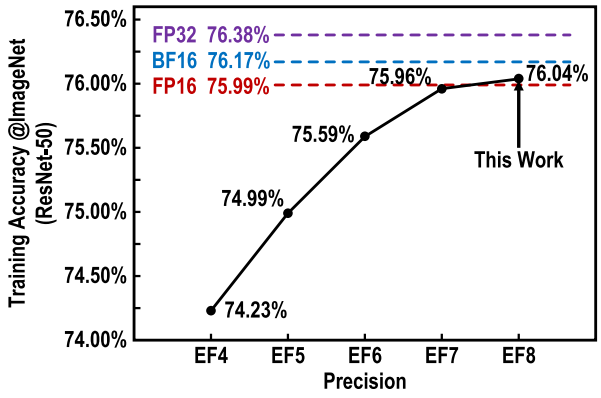


Fig. 24. Exponent-exclusive precision vs. accuracy trade-off and comparison with FP16/BF16 baselines of ResNet-50 on ImageNet.

to 0.95 V. The MVM CIM core achieves an energy efficiency of 28.5-99.0 TFLOPS/W and an area efficiency of 1.02-3.94 TFLOPS/mm^2 for EBP. The highest energy efficiency of 99.0 TFLOPS/W is measured at the minimum supply voltage of 0.55 V. The VOP CIM core achieves an energy efficiency of 37.2-115.9 TFLOPS/W and an area efficiency of 0.96-3.72 TFLOPS/mm^2 for fused WGC and WU. Thanks to the hardware-efficient EF8 processing of the learning rate in WU, the throughput of fused WGC and WU in the VOP CIM core is higher than EBP operation in the MVM CIM core. The shmoo plot is presented in Fig. 21. By leveraging the sparsity-aware addend-compression adder tree and adder-free SRAM signed ripple counter, the access time is up to 2.4 ns at 0.95 V supply voltage.

To evaluate the training accuracy, a TEF-CIM-based training platform is established. In this platform, the TEF-CIM macro is employed to accelerate the training operations of general matrix multiplication in CNNs and Transformers, including forward propagation, EBP, WGC, and WU. The training flow for convolution, linear, and attention layers is depicted in Fig. 22. The normalization and activation training accelerator can be designed in hardware efficiently, like in [39]. Fig. 23 shows the training accuracy curve of the TEF-CIM-based training platform for ResNet-50 on ImageNet, including the impact of the approxmiate computation in the MVM CIM core. Compared to the FP32 software baseline, this work achieves stable convergence with a training accuracy loss of 0.34%. Fig. 24 illustrates that as the precision increases from EF4 to EF8, the training accuracy of ResNet-50 on ImageNet improves from 74.23% to 76.04%, demonstrating stable convergence with increasing dynamic range. Under the same training configuration, including the same optimizer and training schedule, the proposed EF8 achieves 76.04% training accuracy, which is 0.34% lower than the FP32 baseline and 0.14% lower than the BF16 baseline due to the absence of mantissa. Nevertheless, EF8 achieves 0.05% higher accuracy than FP16 with its larger dynamic range.

By performing WGC and WU with high precision in the VOP CIM core with initial FP16 weight rather than EF8 weight, the optimization stability of the training framework is maintained, as shown in Fig. 23. To guarantee training convergence, this work adopts quantization-aware training based on the Straight-Through Estimator, which allows gradients to directly pass through the non-differentiable quantizers of activations and weights during the backward pass [15], [16]. In addition, the SGD optimizer employs a multi-stage learning-rate scheduler with learning rates of 2^{-3}, 2^{-7}, and 2^{-10} for different training stages.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE II
PERFORMANCE COMPARISON TABLE

	This Work	JSSC'25 [11]	JSSC'22 [13]	JSSC'24 [32]	JSSC'23 [8]	ISSCC'25 [24]	TCAS-I'26 [28]
Technology	28nm	28nm	28nm	28nm	28nm	28nm	28nm
Computation Domain	Digital	Digital	Analog	Digital	Digital	Digital	Digital
Operation	**FWP/EBP/WGC/WU** [a]	FWP [a]	FWP/EBP [a]	FWP [a]	FWP/EBP/WGC [a]	FWP [a]	FWP [a]
Application	**CNN/Transformer**	CNN/Transformer	CNN	CNN	CNN	CNN/Transformer	CNN
Memory Type of CIM	**Comp. 5T-SRAM**, 8T-SRAM	8T-SRAM	6T-SRAM	6T-SRAM	8T-SRAM	10T-SRAM	6T-SRAM
Data Format	**EF8 @Error, IA, W (EBP)** **FP16 @W (WU)**	INT1-16, FP16, BF16, FP4/6/8	INT8	INT8, BF16	INT1-16	INT8, FP16, FP8	INT1-8
Norm. Ext. Mem. Access Reduction [b]	**63%**	-	33% (INT8)	0% (BF16)	24% (INT16)	33% (INT8/FP8)	0% (BF16)
Voltage (V)	0.55-0.95	0.65-0.9	0.85-1	0.6-0.9	0.75-1.05	0.62-0.9	0.6-1.1
Frequency (MHz)	110-425	60-300	-	-	50-280	100-525	30-360
Peak Area Efficiency (TFLOPS/mm^2 or TOPS/mm^2)	**3.94 (EBP)** **3.72 (WGC & WU)** **3.81 (EBP & WGC & WU)**	0.56 @FP8	-	2.57 @INT8 2.05 @BF16	0.39 @INT16	3.81 @INT8 2.44 @FP16 3.30 @ FP8	1.43 @INT8
Energy Efficiency (TFLOPS/W or TOPS/W)	**28.5-99.0 (EBP)** **37.2-115.9 (WGC & WU)** **33.1-108.5 (EBP & WGC & WU)**	10.2-13.3 @FP8	7-7.6 @INT8	19.5-44 @INT8 14.04-31.6 @BF16	0.84-7.59	14.2-49.6 @INT8 4.8-16.2 @FP16 11.7-31.2 @FP8	22.4-60.4 @INT8
Inference/Training Accuracy	91.60% @CIFAR10 (ResNet-20) 77.48% @CIFAR100 (ResNet-18) 76.04% @ImageNet (ResNet-50) **88.13%% @SQuAD (BERT-base)**	68.5% @ImageNet (ResNet-18) 75.2% @ImageNet (ResNet-50) 80.6% @ImageNet (ViT-B)	91.94% @CIFAR10 (ResNet-20) 67.79% @CIFAR100 (ResNet-20)	75.94% @CIFAR100 (ResNet-18)	90.07% @CIFAR10 (VGG-16) 61.68% @CIFAR100 (VGG-16)	76.30% @ImageNet (ResNet-50)	-
Inference/Training Accuracy Loss	**0.20% @CIFAR10 (ResNet-20)** **0.28% @CIFAR100 (ResNet-18)** **0.34% @ImageNet (ResNet-50)** **0.23% @SQuAD (BERT-base)**	-	0.49% @CIFAR10 (ResNet-20) 0.65% @CIFAR100 (ResNet-20)	0.17% @CIFAR100 (ResNet-18)	-	0.1% @ImageNet (ResNet-50)	-

[a] FWP: forward propagation [b] External memory access amount reduction form MVM FP16 CIM in ResNet-50 training (batch size=8)

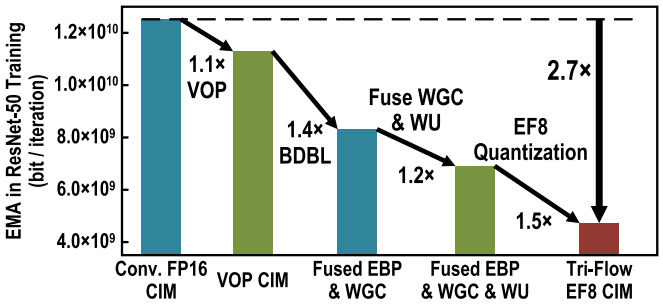


Fig. 25. External memory access.

Fig. 25 compares external memory access across five CIM-based training platform configurations for ResNet-50 training on ImageNet (batch size = 8). The proposed VOP CIM core reduces external memory access by 1.1× compared to typical FP16 CIM. The fusion of EBP, WGC, and WU achieves a 1.9× reduction of external memory access relative to previous CIM architectures. The exponent-exclusive arithmetic of EF8 achieves an additional 1.5× reduction. By integrating all the proposed techniques, the total external memory access is reduced by 2.7×.

Table II summarizes the performance comparison between this work and the SOTA. The proposed TEF-CIM macro supports the most comprehensive training pipeline (forward propagation, EBP, WGC, and WU) and uniquely extends to CNN and Transformer. This work achieves the highest area efficiency of 3.81 TFLOPS/mm^2, which is closely related to the training time. The area efficiency is 1.48× higher than the SOTA INT8 CIM. For the energy efficiency of fused EBP, WGC, and WU (mean value of two CIM cores' energy efficiency), this work achieves 1.80× and 3.43× higher performances than the SOTA INT8/BF16 CIM. Prior FP8 CIM work only supports inference. Compared with [11] and [24], the MVM CIM core achieves more than 1.19× higher area efficiency and 3.17× higher energy efficiency. The training accuracy drops for ResNet-20, ResNet-18, ResNet-50, and BERT-Base are 0.20%, 0.28%, 0.34%, and 0.23%, respectively, including the accuracy loss caused by approximate computation in the MCM CIM core. Since the TEF-CIM-based training platform employs EF8 quantization for both forward propagation and backpropagation during training, the same EF8 computation framework can also be directly applied to inference deployment. Therefore, the reported converged training accuracies inherently reflect the inference accuracies evaluated on the corresponding test datasets after training convergence.

VI. CONCLUSION

This article presented a TEF-CIM macro in 28-nm CMOS. Due to the heterogeneous tri-flow data loading system and exponent-exclusive EF8 arithmetic, the design fuses EBP, WGC, and WU with energy-efficient hardware, which reduces the external memory access by 2.7× compared to conventional FP16 CIM macros. By leveraging the uniqueness of the exponent-exclusive format in the proposed bit-polarity-interleaved addend-compression MVM CIM block and the SRAM bit-select ripple-counter-based VOP cell, the TEF-CIM macro achieves an energy efficiency of 108.5 TFLOPS/W and an area efficiency of 3.81 TFLOPS/mm^2, which are more than 1.80× and 1.48× higher than the SOTA INT8/BF16 CIM macros.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

REFERENCES

[1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 28492–28518.

[2] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.

[3] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 1, p. 140, 2020.

[4] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, no. 1, pp. 99–106, Jan. 2022.

[5] Y. Yang, G. Li, and R. Marculescu, "Efficient on-device training via gradient filtering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3811–3820.

[6] J. Lin, L. Zhu, W.-M. Chen, W.-C. Wang, C. Gan, and S. Han, "On-device training under 256KB memory," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 22941–22954.

[7] Y. Wang et al., "Trainer: An energy-efficient edge-device training processor supporting dynamic weight pruning," *IEEE J. Solid-State Circuits*, vol. 57, no. 10, pp. 3164–3178, Oct. 2022.

[8] J. Heo, J. Kim, S. Lim, W. Han, and J.-Y. Kim, "T-PIM: An energy-efficient processing-in-memory accelerator for end-to-end on-device training," *IEEE J. Solid-State Circuits*, vol. 58, no. 3, pp. 600–613, Mar. 2023.

[9] F. Tu et al., "ReDCIM: Reconfigurable digital computing- in -memory processor with unified FP/INT pipeline for cloud AI acceleration," *IEEE J. Solid-State Circuits*, vol. 58, no. 1, pp. 243–255, Jan. 2023.

[10] S. Yan et al., "A 28-nm floating-point computing-in-memory processor using intensive-CIM sparse-digital architecture," *IEEE J. Solid-State Circuits*, vol. 59, no. 8, pp. 2630–2643, Aug. 2024.

[11] Y. Zhan et al., "AFP-CIM: All-inclusive floating-point with segmented compute-in-memory macro," *IEEE J. Solid-State Circuits*, vol. 61, no. 6, pp. 3018–3031, Jun. 2026.

[12] Y. Yuan et al., "14.5 a 28nm 192.3TFLOPS/W accurate/approximate dual-mode-transpose digital 6T-SRAM CIM macro for floating-point edge training and inference," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2025, pp. 258–260.

[13] J.-W. Su et al., "Two-way transpose multibit 6T SRAM computing-in-memory macro for inference-training AI edge chips," *IEEE J. Solid-State Circuits*, vol. 57, no. 2, pp. 609–624, Feb. 2022.

[14] C. Zhao et al., "A 28-nm 36 kb SRAM CIM engine with 0.173 μm2 4T1T cell and self-load-0 weight update for AI inference and training applications," *IEEE J. Solid-State Circuits*, vol. 59, no. 10, pp. 3277–3289, Oct. 2024.

[15] B. Chmiel, R. Banner, E. Hoffer, H. B. Yaacov, and D. Soudry, "Accurate neural training with 4-bit matrix multiplications at standard formats," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023, pp. 1–20.

[16] J. Zhao et al., "LNS-madam: Low-precision training in logarithmic number system using multiplicative weight update," *IEEE Trans. Comput.*, vol. 71, no. 12, pp. 3179–3190, Dec. 2022.

[17] F. Abecassis et al., "Pretraining large language models with NVFP4," 2025, *arXiv:2509.25149*.

[18] P. Haghi et al., "Bridging the gap between LLMs and LNS with dynamic data format and architecture codesign," in *Proc. 57th IEEE/ACM Int. Symp. Microarchitecture (MICRO)*, Nov. 2024, pp. 1617–1631.

[19] S. Vogel, M. Liang, A. Guntoro, W. Stechele, and G. Ascheid, "Efficient hardware acceleration of CNNs using logarithmic data representation with arbitrary log-base," in *Proc. IEEE/ACM Int. Conf. Comput.-Aided Design (ICCAD)*, Nov. 2018, pp. 1–8.

[20] C. Liu et al., "Ultra-low precision multiplication-free training for deep neural networks," 2023, *arXiv:2302.14458*.

[21] X. Sun et al., "Ultra-low precision 4-bit training of deep neural networks," in *Proc. NIPS*, vol. 33, 2020, pp. 1796–1807.

[22] Y. Fu et al., "FLEX-CIM: A flexible kernel size 1-GHz 181.6-TOPS/W 25.63-TOPS/mm^2 analog compute-in-memory macro," *IEEE J. Solid-State Circuits*, vol. 59, no. 9, pp. 3021–3031, Sep. 2024.

[23] X. Xiang, Z. Yue, X. Zhang, S. Wei, Y. Hu, and S. Yin, "Dyn-bitpool: A 28 nm 27 TOPS/W two-sided sparse CIM accelerator featuring a balanced workload scheme and high CIM macro utilization," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 5, pp. 2216–2228, May 2025.

[24] Z. Yue et al., "14.4 A 51.6TFLOPs/W full-datapath CIM macro approaching sparsity bound and $< 2^{-30}$ loss for compound AI," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2025, pp. 1–3.

[25] A. Guo et al., "A 28 nm 128-kb exponent- and mantissa-computation-in-memory dual-macro for floating-point and INT CNNs," *IEEE J. Solid-State Circuits*, vol. 60, no. 10, pp. 3639–3654, Oct. 2025.

[26] Y. He et al., "An area-efficient lookup-table -based eDRAM digital CIM macro for neural network inference," *IEEE J. Solid-State Circuits*, vol. 61, no. 4, pp. 1697–1709, Apr. 2026.

[27] H. You, W. Li, D. Shang, Y. Zhou, and S. Qiao, "A 1–8b reconfigurable digital SRAM compute-in-memory macro for processing neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 4, pp. 1602–1614, Apr. 2024.

[28] J. Oh, C.-T. Lin, and M. Seok, "D6CIM: 60.4-TOPS/W all-digital 6T-SRAM-based compute-in-memory macro supporting 1-to-8 b fixed-point arithmetic in a 28-nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 73, no. 7, pp. 4789–4800, Jul. 2026.

[29] Y. Yuan et al., "34.6 A 28nm 72.12TFLOPS/W hybrid-domain outer-product based floating-point SRAM computing-in-memory macro with logarithm bit-width residual ADC," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 576–578.

[30] A. Guo et al., "34.3 A 22nm 64kb lightning-like hybrid computing-in-memory macro with a compressed adder tree and analog-storage quantizers for transformer and CNNs," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 570–572.

[31] S. Jung et al., "A hybrid digital-analog compute-in-memory using content-addressable memory with flexible multi-bit slicing," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 73, no. 5, pp. 3452–3465, May 2026.

[32] A. Guo et al., "A 28-nm 64-kb 31.6-TFLOPS/W digital-domain floating-point-computing-unit and double-bit 6T-SRAM computing-in-memory macro for floating-point CNNs," *IEEE J. Solid-State Circuits*, vol. 59, no. 9, pp. 3032–3044, Sep. 2024.

[33] V. Sharma, X. Zhang, N. S. Dhakad, and T. T.-H. Kim, "FlexDCIM: A 400 MHz 249.1 TOPS/W 64 kb flexible digital compute-in-memory SRAM macro for CNN acceleration," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 10, pp. 5696–5707, Oct. 2025.

[34] Y.-D. Chih et al., "16.4 an 89TOPS/W and 16.3TOPS/mm^2 all-digital SRAM-based full-precision compute-in memory macro in 22nm for machine-learning edge applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2021, pp. 252–254.

[35] J. Wang et al., "A 28-nm compute SRAM with bit-serial logic/arithmetic operations for programmable in-memory vector computing," *IEEE J. Solid-State Circuits*, vol. 55, no. 1, pp. 76–86, Jan. 2020.

[36] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm^2 fully-digital computing-in-memory macro supporting wide-range dynamic-voltage-frequency scaling and simultaneous MAC and write operations," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 65, Feb. 2022, pp. 1–3.

[37] Y. Zhan, W.-H. Yu, K.-F. Un, R. P. Martins, and P.-I. Mak, "GSLP-CIM: A 28-nm globally systolic and locally parallel CNN/transformer accelerator with scalable and reconfigurable eDRAM compute-in-memory macro for flexible dataflow," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 4, pp. 1657–1667, Apr. 2025.

[38] H. Zhang et al., "A 90.7-nW vibration-based condition monitoring chip featuring a digital compute-in-memory- based DNN accelerator using an ultra-low-power 13T-SRAM cell," *IEEE J. Solid-State Circuits*, vol. 60, no. 1, pp. 321–331, Jan. 2025.

[39] D. Han et al., "HNPU: An adaptive DNN training processor utilizing stochastic dynamic fixed-point and active bit-precision searching," *IEEE J. Solid-State Circuits*, vol. 56, no. 9, pp. 2858–2869, Sep. 2021.



Zhenyu Wang (Student Member, IEEE) received the B.S. degree in electronic science and engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2022. He is currently pursuing the Ph.D. degree with the Faculty of Science and Technology and the State Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), Department of Electrical and Computer Engineering, University of Macau, Macau, China.

His research interests include reconfigurable compute-in-memory circuit designs and edge AI training accelerator designs.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

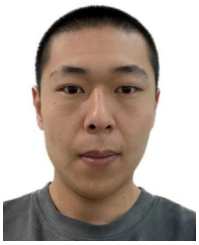
This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.



Yongqiang Mo received the B.S. degree in integrated circuit design from Guangdong University of Technology (GDUT), Guangzhou, China, in 2024, and the M.S. degree in microelectronics from the University of Macau (UM), Macau, China, in 2026. He is currently a Chip Design Engineer with Huawei Technologies Company Ltd. His research interests include artificial intelligence neural network accelerator design, digital chip physical design, and backend timing optimization.



Junren Chen received the B.Sc. and M.Sc. degrees in electrical and computer engineering and microelectronics from the University of Macau (UM), Macau, China, in 2024 and 2026, respectively. He was a member of the State-Key Laboratory of Analog and Mixed-Signal VLSI, UM, where he focused on computing-in-memory architectures and systems of artificial intelligence applications. He is currently a Physical Design Engineer with a leading semiconductor company in Shanghai, China.

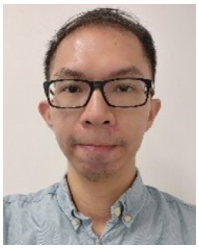


Guoqiang Xin (Student Member, IEEE) received the B.S. degree from the University of Macau (UM), Macau, China, in 2022, where he is currently pursuing the Ph.D. degree with the State Key Laboratory of Analog and Mixed-Signal VLSI (AMSV). He has been with the State Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), UM, since 2021. His research interests include edge AI, in-memory computing, brain–computer interfaces, and signal processing. He received the First Prize in the IEEE Project Competition (IEEE Macau Section) in 2021.



Wei-Han Yu (Member, IEEE) received the Ph.D. degree from the University of Macau (UM), Macau, in 2018.

From 2019 to 2021, he was a Visiting Scholar with the Muramnn Mixed-Signal Group, Stanford University. He has been an Assistant Professor with the State Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau (UM), Macau, China, since 2021. His research interests include edge AI, in-memory computing, switched capacitor circuits, energy-harvested RF transceivers, and neural interfaces. He received the IEEE ISSCC Student Travel Grant Award, the FDCT Science and Technology Postgraduate Student Award in 2016, and the IEEE SSCS Predoctoral Achievement Award in 2018.



Ka-Fai Un (Member, IEEE) received the Ph.D. degree from the University of Macau (UM), Macau, in 2014.

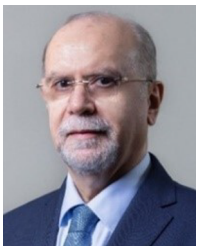
He was a Post-Doctoral Researcher and a Lecturer (Macau Fellow) with UM in 2014 and 2015, respectively. He was on leave from UM and was a Visiting Post-Doctoral Researcher with the School of Electrical and Electronic Engineering, University College Dublin, Dublin, Ireland, from 2017 to 2018. He has been an Assistant Professor with the State Key Laboratory of Analog and Mixed-Signal VLSI, UM, since 2018. He is currently an Associate Professor. His research interests include analog artificial intelligence (AI) circuits, digital AI accelerator design, and analog and RF circuit design. In 2003, he received Macau Mathematics Olympiad and represented Macau in Chinese Mathematics Olympiad (CMO) and the International Mathematics Olympiad (IMO), in Changsha and Tokyo, respectively. He was a recipient of the 2008 APCCAS Merit Student Paper Certificate. He has been an Associate Editor of IEEE SOLID-STATE CIRCUITS LETTERS since 2024.



Jiawei Xu (Member, IEEE) received the Ph.D. degree in microelectronics from Fudan University, Shanghai, China, in 2022. She was a Post-Doctoral Researcher with the KTH Royal Institute of Technology, Sweden. She has been an Assistant Professor with the State Key Laboratory of Analog and Mixed-Signal VLSI (AMSV), University of Macau, since 2024. Her current research interests include deep compression, energy-efficient neural network accelerators, neuromorphic computing architecture, and brain-inspired chip design.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.



Rui P. Martins (Life Fellow, IEEE) was born in April 1957. He received the bachelor's, master's, Ph.D., and Habilitation (for Full-Professor) degrees in electrical engineering and computers from the Department of Electrical and Computer Engineering (DECE), Instituto Superior Técnico (IST), University of Lisbon, Portugal, in 1980, 1985, 1992, and 2001, respectively.

He has been with the DECE/IST, University of Lisbon, since October 1980. Since October 1992, has been on leave from the University of Lisbon and with the DECE, Faculty of Science and Technology (FST), University of Macau (UM), Macau, China, where he has been the Chair-Professor since August 2013. In FST, he was the Dean from 1994 to 1997, and has been an UM's Vice-Rector since September 1997. From September 2008 to August 2018, he was the Vice-Rector (Research). From September 2018 to August 2023, he was the Vice-Rector (Global Affairs). He created 2003, the Analog and Mixed-Signal VLSI Research Laboratory, UM, elevated in January 2011, to the State Key Laboratory (SKLAB) of China (the first in engineering in Macau), being its Founding Director. He was the Founding Chair of UMTEC (UM company) from January 2009 to March 2019, supporting the incubation and creation; and Digifluidic, the first UM Spin-Off, in 2018, whose CEO is a SKLAB Ph.D. Graduate. He was also the Co-Founder of Chipidea Microelectronics (Macau) [later Synopsys-Macau, and now Akrostar, from 2001 to 2002, where the CEO is one of his Ph.D. graduates]. Within the scope of his teaching and research activities he has taught 21 bachelor's and master's courses. In UM, he has supervised (or co-supervised) 47 theses, Ph.D. (26), and master's (21). He has authored or co-authored: nine books and 12 book chapters; 49 Patents, USA (39), Taiwan (three), and China (seven); 675 articles, in scientific journals (289) and in conference proceedings (386); and other 70 academic works, in a total of 815 publications.

Prof. Martins is an IEEE Fellow. He is a member of the Advisory Board of the Journal of Semiconductors of the Chinese Institute of Electronics (CIE), the Institute of Semiconductors, and Chinese Academy of Sciences, since January 2021, and a fellow of Asia–Pacific Artificial Intelligent Association since October 2021. He was also a member of the IEEE CASS Fellow Evaluation Committee in 2013, 2014, and 2018, respectively, where he was the Chair, in 2019, and the Vice-Chair in 2021 and 2022, respectively; the IEEE Nominating Committee of Division I Director (CASS/EDS/SSCS) in 2014; and the IEEE CASS Nominations Committee from 2016 to 2017. He was the IEEE Council on Electronic Design Automation (CEDA) Outstanding Service Award in 2016, and also the General Chair of the IEEE Asian Solid-State Circuits Conference (A-SSCC) in 2019. He was the Founding Chair of the IEEE Macau Section from 2003 to 2005 and the IEEE Macau Joint-Chapter on Circuits And Systems (CAS)/Communications (COM) from 2005 to 2008, where received the 2009 World Chapter of the Year of IEEE CAS Society (CASS). He served as the General Chair of the IEEE Asia–Pacific Conference on CAS (APCCAS 2008), the Vice-President (VP) Region 10 (Asia, Australia, and Pacific) from 2009 to 2011 and as the VP-World Regional Activities and Membership of IEEE CASS from 2012 to 2013. In addition, he was the General Chair of the ACM/IEEE Asia South Pacific Design Automation Conference (ASP-DAC 2016). He was the Vice-President from 2005 to 2014, the President from 2014 to 2017, and the Vice-President from 2021 to 2024 of the Association of Portuguese Speaking Universities (AULP). He received three Macau Government decorations: the Medal of Professional Merit (Portuguese) in 1999; the Honorary Title of Value (Chinese) in 2001; and the Medal of Merit in Education (Chinese) in 2021. Since July 2010, he has been elected, unanimously, to the Lisbon Academy of Sciences, where he served as a Corresponding from 2010 to 2022 and an Effective Member in 2022. He is currently the only Portuguese academician working and living in Asia. He was also an Associate-Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS from 2010 to 2013. He was nominated the Best Associate Editor from 2012 to 2013.



Pui-In Mak (Fellow, IEEE) received the Ph.D. degree from the University of Macau (UM), Macau, China, in 2006.

He is currently a Full Professor with the Faculty of Science and Technology, ECE Department, UM; the Interim Director with the State Key Laboratory of Analog and Mixed-Signal VLSI; and the Deputy Director (Research) with the Institute of Microelectronics. His research interests include analog and radio-frequency (RF) circuits and systems for wireless and multidisciplinary innovations.

Prof. Mak has been inducted as an Overseas Expert of Chinese Academy of Sciences since 2018; a fellow of U.K. Institution of Engineering and Technology (IET) for contributions to engineering research, education, and services, since 2018; a fellow of IEEE "for contributions to radio-frequency and analog circuits" since 2019, and a fellow of U.K. Royal Society of Chemistry since 2020. He (co)-received the DAC/ISSCC Student Paper Award in 2005, the CASS Outstanding Young Author Award in 2010, the National Scientific and Technological Progress Award in 2011, the A-SSCC Distinguished Design Award in 2015, and the ISSCC Silkroad Award in 2016. He was decorated with the Honorary Title of Value for scientific merits by Macau Government. He is/was the TPC Vice Co-Chair of ASP-DAC in 2016, and a TPC Member of A-SSCC from 2013 to 2016 and in 2019, ESSCIRC from 2016 to 2017, and ISSCC from 2017 to 2019. He is/was a Distinguished Lecturer of the IEEE Circuits and Systems Society from 2014 to 2015 and the IEEE Solid-State Circuits Society from 2017 to 2018. He was the Chairperson of the Distinguished Lecturer Program of IEEE Circuits and Systems Society from 2018 to 2019. His involvements with IEEE are: an Editorial Board Member of IEEE Press from 2014 to 2016; a Member of Board-of-Governors of the IEEE Circuits and Systems Society from 2009 to 2011; a Senior Editor of IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS from 2014 to 2015; an Associate Editor of IEEE JOURNAL OF SOLID-STATE CIRCUITS since 2018, IEEE SOLID-STATE CIRCUITS LETTERS since 2017, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: EXPRESS BRIEFS from 2010 to 2011 and from 2014 to 15, respectively, and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS from 2010 to 2013. He was the Best Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS from 2012 to 2013.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:29:10 UTC from IEEE Xplore. Restrictions apply.